

Mini Project Report

Title: Rock Paper Scissors Web Game

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Course: INFORMATION TECHNOLOGY

Semester: II

1. Introduction

The Rock-Paper-Scissors game is a simple hand game usually played between two people. Each player simultaneously forms one of three shapes with an outstretched hand: rock, paper, or scissors. In this project, we have recreated this game using HTML, CSS, and JavaScript to allow a user to play against a computer in a web browser.

This project aims to enhance understanding of basic web development concepts and implement game logic through JavaScript in a fun and interactive way.

2. Objective

- To implement a simple Rock-Paper-Scissors game using web technologies.
 - To understand and practice the integration of HTML, CSS, and JavaScript.
 - To explore event-driven programming using JavaScript.
 - To gain experience in creating an interactive front-end application.
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3. Tools and Technologies Used

- **HTML** - For structuring the web page and game elements.

- **CSS** - For styling the page layout and visuals.
 - **JavaScript** - For implementing the core game logic and user interaction.
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4. Pseudocode

INITIALIZE:

```
SET playerScore = 0
SET computerScore = 0
```

DEFINE imagePaths:

```
player:
  rock -> "user-rock.jpg"
  paper -> "user-paper.jpg"
  scissors -> "user-scissor.jpg"
computer:
  rock -> "com-rock.jpg"
  paper -> "com-paper.jpg"
  scissors -> "com-scissor.jpg"
```

// MAIN FUNCTION: Play Game

FUNCTION play(playerChoice):

```
SET choices = ["rock", "paper", "scissors"]
SET computerChoice = RANDOM choice from choices
```

```
SET playerImg = element with id "player-img"
SET computerImg = element with id "computer-img"
```

```
SET playerImg.src = imagePaths.player[playerChoice]
SET computerImg.src = imagePaths.computer[computerChoice]
```

```
DISPLAY both playerImg and computerImg
ADD animation class to both images
```

```
WAIT 400 milliseconds:
  REMOVE animation class from both images
```

```
IF playerChoice == computerChoice THEN
  resultText = "It's a draw!"
ELSE IF playerChoice beats computerChoice THEN
  resultText = "You win! [Explanation]"
  INCREMENT playerScore
```

```
ELSE
    resultText = "You lose! [Explanation]"
    INCREMENT computerScore

UPDATE result text on screen with resultText
UPDATE score display with playerScore and computerScore

// HELPER FUNCTION: Capitalize word
FUNCTION capitalize(word):
    RETURN word with first letter capitalized

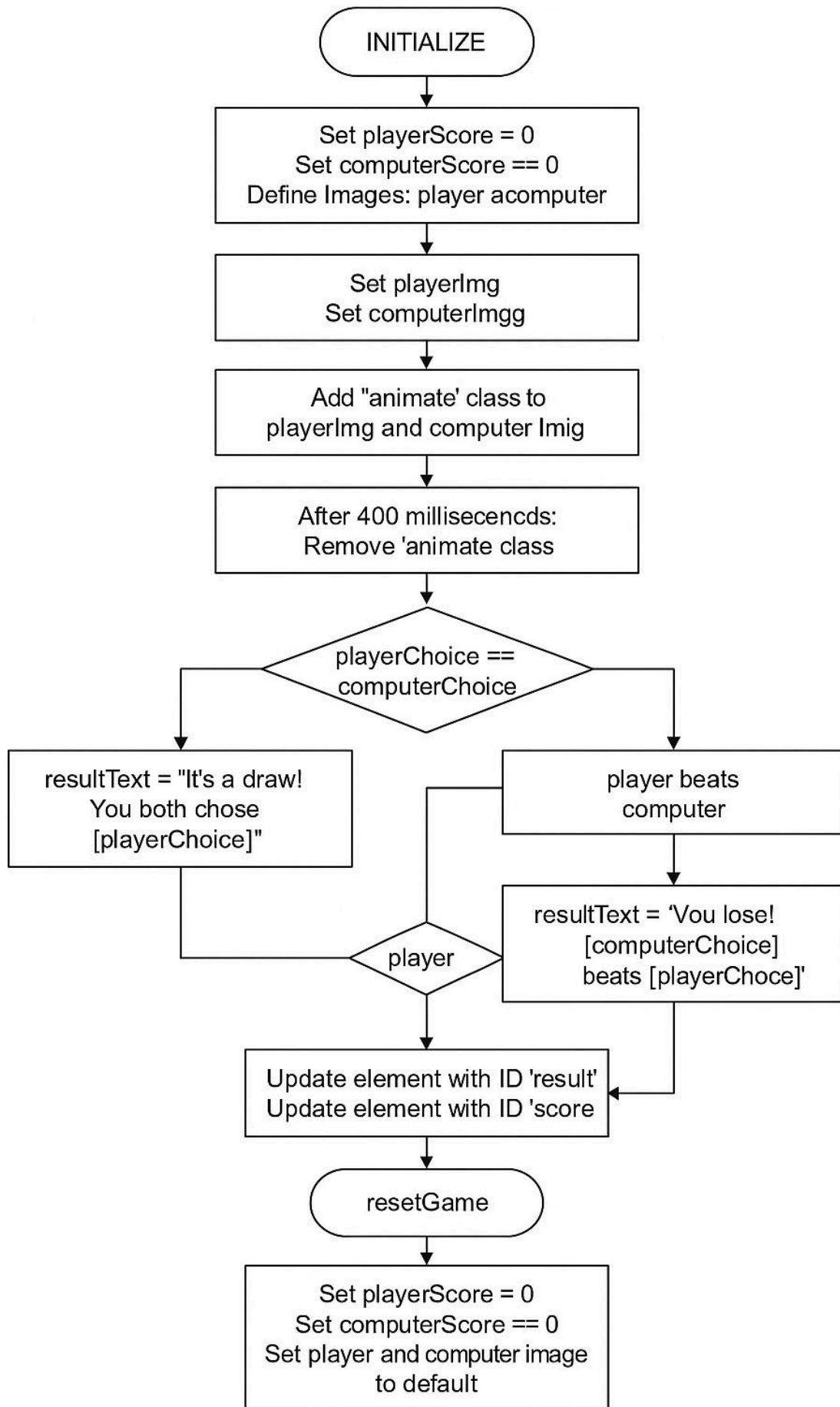
// RESET FUNCTION
FUNCTION resetGame():
    SET playerScore = 0
    SET computerScore = 0
    CLEAR result and score displays
    HIDE playerImg and computerImg
```

5. Flowchart

A flowchart illustrating the game's decision-making process is included below (Insert Image Here).

1. Start
2. Initialize Scores and Image Paths
3. User selects Rock / Paper / Scissors
4. Computer randomly selects a choice
5. Set image sources and apply animation
6. Compare choices:
 - If equal: Draw
 - If user beats computer: User Wins
 - Else: Computer Wins

7. Update result text and scores on screen
8. Repeat or End

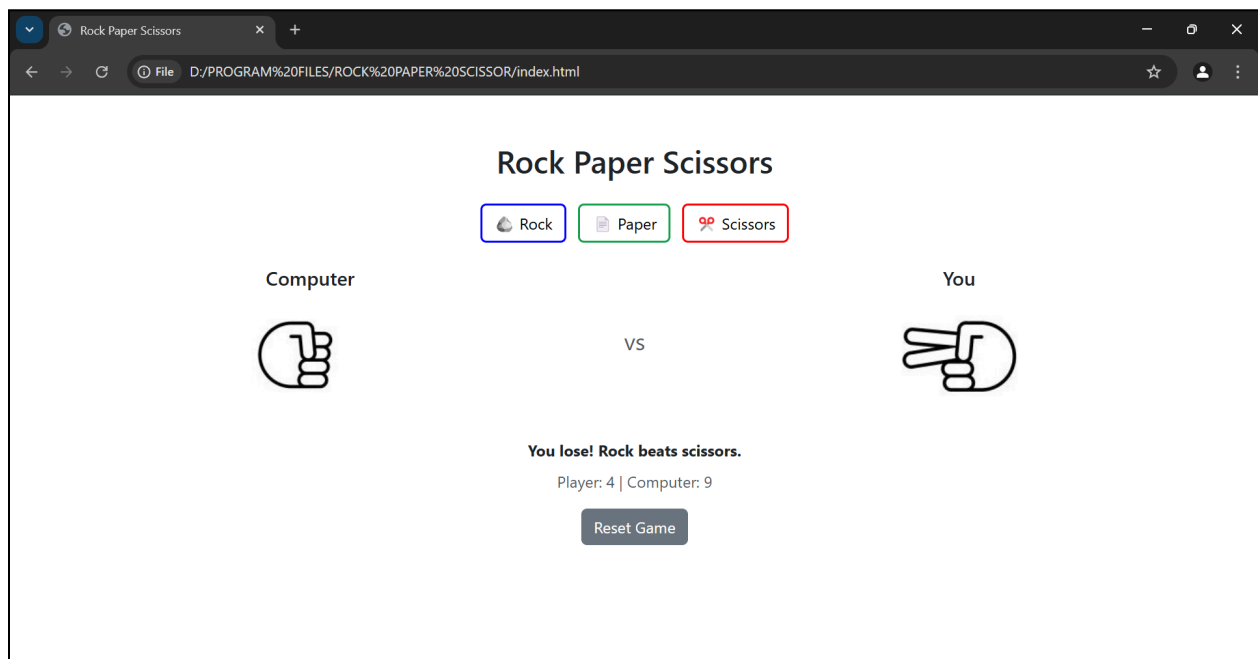
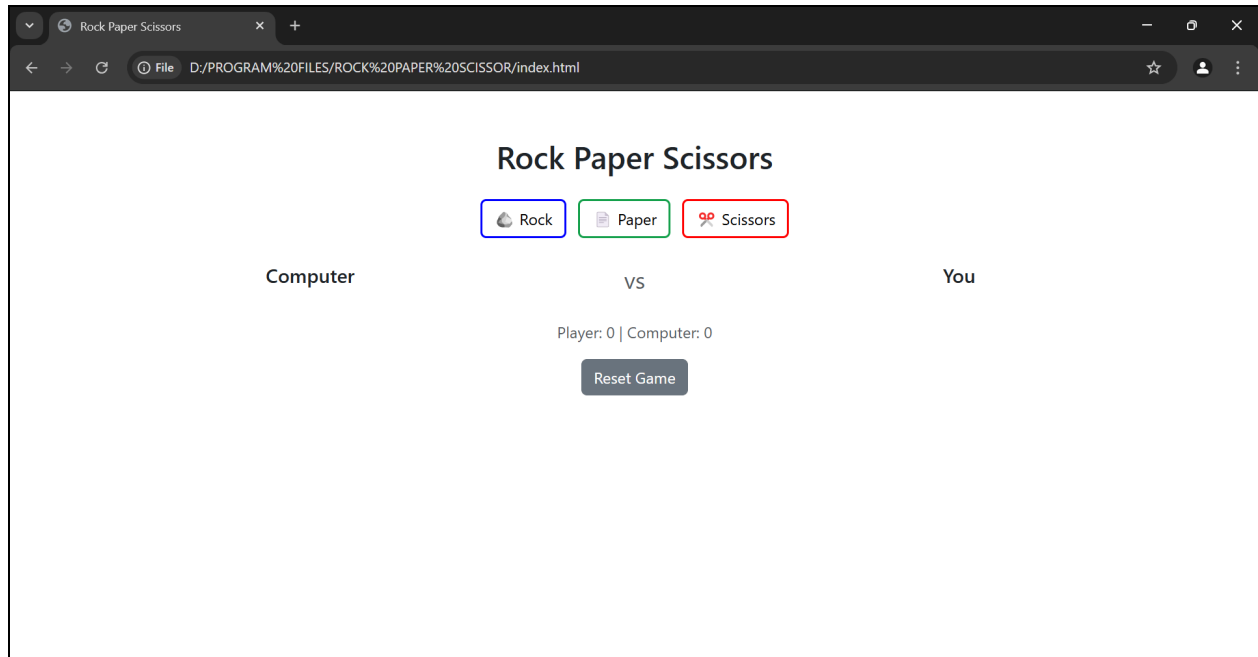


6. Code Overview

A simplified version of the game logic in JavaScript is as follows:

```
function playGame(playerChoice) {  
  const choices = ["rock", "paper", "scissors"];  
  const computerChoice = choices[Math.floor(Math.random() * 3)];  
  
  if (playerChoice === computerChoice) {  
    result = "It's a Tie!";  
  } else if (  
    (playerChoice === "rock" && computerChoice === "scissors") ||  
    (playerChoice === "paper" && computerChoice === "rock") ||  
    (playerChoice === "scissors" && computerChoice === "paper")  
  ) {  
    result = "You Win!";  
  } else {  
    result = "Computer Wins!";  
  }  
  
  displayResult(result);  
}
```

7. Screenshots



8. Conclusion

This mini project provided practical experience in using web technologies to build a browser-based interactive game. It helped reinforce key concepts such as DOM manipulation, event handling, and conditional logic in JavaScript. The game is a functional, enjoyable

demonstration of how simple logic and web development skills can combine to create engaging user experiences.
